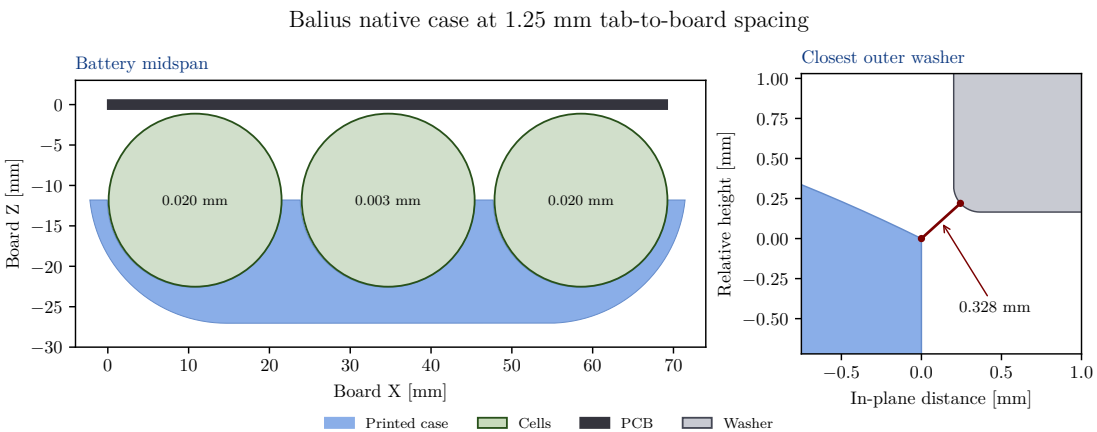
The three-pocket Balius case fits at 1.25 mm stand-off



The arch around the middle cell comes within 0.003 mm. The closest outer washer clears by 0.328 mm. These are rigid distances in the registered SolidWorks geometry.

The shell adds 73.7 g to the 251.8 g bare assembly

Snug geometry

Three cells, four 6.35 mm thick tabs and four M4 case screws. The nominal model uses printed PLA, an air gap, and either nylon or aluminium screws.

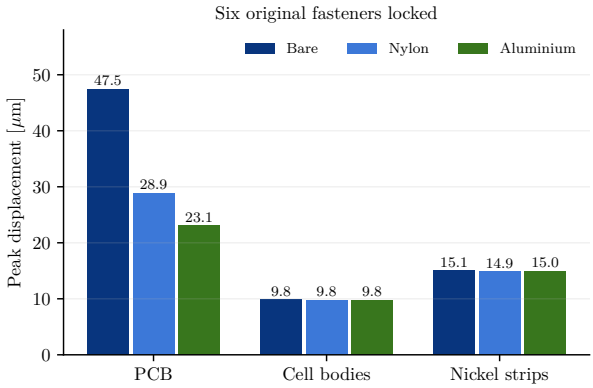
The shell mass assumes dense PLA. Screws and spacer mass are excluded from the estimate.

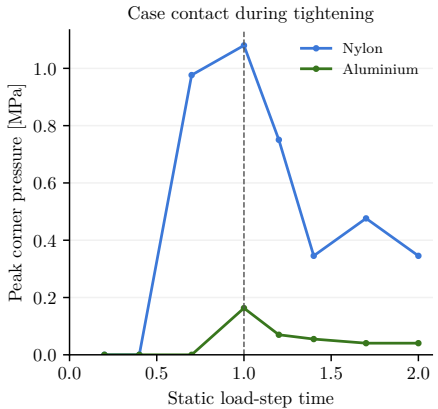
No-case reference

First prestressed mode 485.96 Hz. Peak board-normal motion under base Z was 52.67 μm RMS. At 0.50 mm lateral travel the bare fixture required 8.54 N in positive X and 8.51 N in negative X.

The rerun compares the same loads and movie subjects. The earlier flush and 3 mm results are preserved in Appendix A of the report.

Both snug mounts reduce PCB movement during tightening





At bolt lock, peak PCB displacement fell from **47.46 μm bare** to **28.91 with nylon screws** and **23.09 with aluminium screws**. Only the middle cell contacted the case. Both outer cells and all six washers stayed clear.